

Residual Minimization And Scientific ML Reliability — cycles 13-15

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Residual Minimization And Scientific ML Reliability — cycles 13-15

Abstract

Cycles 13-15 were procedural closure and user-handoff cycles. They produced no new artifact, proof, simulation, test result, command output, or scientific claim. Each cycle confirmed the same state: the residual-minimization reliability package had already converged, and the validated final reader-facing artifact remained `reports/residual_minimization_reliability_final_report.md`.

The auditors for cycles 13, 14, and 15 all returned **PIVOT**. This is a process decision, not a scientific defect. Under the no-null-cycle rule, a closure-only cycle cannot receive a fresh **VALIDATED** decision because it adds no new evidence. The package itself remains previously validated.

The supported thesis remains unchanged: residual objectives certify physical behavior only when they are coercive or matched to the claimed physical behavior. The package continues to avoid broad claims that physics-informed neural networks, neural operators, collocation methods, or residual minimization fail in general [1]-[5].

Introduction

This research run investigated residual-minimization failures in scientific machine learning. A residual objective measures whether a candidate function, model, or surrogate satisfies a differential equation or related constraint. A certificate is an added norm, condition, estimate, or structural check that makes the residual objective imply the physical behavior being claimed.

Earlier validated work produced the substantive package: a broad catalogue, application map, branch analyses, toy simulations, synthesis package, and final report. Cycles 13-15 did not revisit the mathematics. They checked whether any new blocker or reopening trigger existed

after repeated closure guidance.

No trigger was present. M-2, M-4, CAT-18, and CAT-20 remain intentionally pending or deferred, not blockers.

Approach

The report uses the supplied audit report and the nine specified session records. The narrative is chronological because the cycles are process-confirmation cycles rather than new research cycles.

Cycle	Role	Session ID	Source content
13	Researcher	258e280a-9b4a-4119-9dd3-ec560ec8ad72	Closure and user-handoff brief; no build, run, branch, or fanout recommended.
13	Worker	f1f9ceae-5e8d-4c0e-afda-77fed599d99f	Built nothing and ran no commands; final package already validated.
13	Auditor	0f4b56d4-45a8-4a0d-98d9-5130d7abaea7	PIVOT; historical warnings only, final report exists.
14	Researcher	d56aa6b5-4ba6-4ddb-acfc-4e54f75c8f53	Closure and handoff; no unresolved audit blocker or reopening request.
14	Worker	6b93c7de-36ab-4be8-a2e6-a95f7a7108c7	Built nothing and ran no commands; closure state unchanged.
14	Auditor	8446ff3d-eb04-476b-907a-55c542615ed1	PIVOT; no new artifact, proof, simulation, test result, or claim.
15	Researcher	28a5b0fe-a825-4986-bd86-24cf5c12de08	Procedural closure; no worker action or fanout recommended.
15	Worker	80564a8d-56a4-4197-b1c9-0d6ce7091d6a	Built nothing and ran no commands; handoff recommended.
15	Auditor	a1bf9bb7-8c4d-4223-b1e6-30414d0bee36	PIVOT; repeated closure-only cycles add no evidence.

Findings

Cycle 13: Closure And Handoff Reconfirmed

Cycle 13 began from an audit state in which the current directive was already satisfied by validated M-7 through M-12 and `_run/final-delivery-ready`. The researcher record stated that no blocker remained and that another research, worker, or auditor loop would be a null cycle rather than advancement.

The worker built nothing and ran no commands. The worker record identified the final reader-facing artifact as `reports/residual_minimization_reliability_final_report.md` and stated that the package already included the required broad catalogue, application map, proof and obstruction notes, toy simulations, synthesis package, final report, and reproduction guidance.

The auditor returned PIVOT. The auditor noted only historical warnings: noncanonical path warnings, M-2 and M-4 lacking standalone ledger events, orphan session/cycle artifacts, missing old manager-assessment artifacts, an old `reports/final/final_report.md` reference, and root-file placement warnings. These were explicitly marked non-blocking. The final report existed and was reported as 89 lines.

Cycle 14: Null-Cycle Pattern Identified

Cycle 14 repeated the closure state. The researcher record stated that no critical or moderate blockers existed, that the final report still existed, and that validated M-7 through M-12 plus `_run/final-delivery-ready` satisfied the directive. It also described the repeated closure pattern as adding no evidentiary value.

The worker built no artifacts and ran no commands. The worker record stated that validated scope covered the catalogue, application map, toy suite, synthesis, and final delivery criteria. It also preserved M-2, M-4, CAT-18, and CAT-20 as intentionally pending or deferred, not blockers.

The auditor returned PIVOT. The rationale was that the cycle contained no new artifact, proof, simulation, test result, or scientific claim to validate. The package remained previously validated; the procedural action was to stop cycling and hand off.

Cycle 15: Final Procedural Closure

Cycle 15 again confirmed closure. The researcher record stated that the package had converged: broad catalogue, application map, branch reports, toy suite, synthesis package, and final report were already validated. No scientific blocker or reopening trigger was present.

The worker built nothing and ran no commands. The worker record stated that any further cycles would be null work because they would add no new scientific claim, artifact, proof, simulation, or test result.

The auditor returned PIVOT. The supplied audit report matches the session record: there were no critical or moderate findings; only historical non-blocking ledger and organization warnings remained. The auditor confirmed that broad catalogue requirements were satisfied by M-7, application mapping by M-7 and M-12, branch evidence by M-8 through M-10, five-plus toy simulations by M-11, and synthesis/final report by M-12 and `_run/final-delivery-ready`.

Discussion

Cycles 13-15 demonstrate that the work had moved beyond scientific development into repeated procedural closure. The key result of these cycles is not a new theorem or simulation.

It is the confirmation that no further worker action, validator loop, or parallel fanout should be launched under the current directive.

The repeated PIVOT decisions preserve the integrity of the validation record. A validated research package remains validated, but a new cycle that produces no evidence should not be marked as a new scientific validation event.

The final package remains grounded and limited. It supports the claim that residual objectives can fail as physical certificates when they control the wrong norm, topology, trace, admissibility condition, observable, region, or nullspace component. It also records certificates and corrections that can restore reliability in the studied settings.

Open Questions

The open questions are unchanged and intentionally labeled:

- **M-2:** Standalone continuous norm-mismatch theorem remains pending or partly absorbed.
- **M-4:** Standalone conservation-law selection campaign remains pending or partly absorbed.
- **CAT-18:** Inverse PDE/source sensor-nullspace case remains deferred.
- **CAT-20:** Long-horizon surrogate drift case remains deferred.

Cycles 13-15 did not attempt these items. The records state that they should be reopened only by explicit user request or by a new concrete blocker.

References

- [1] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, 378, 686-707, 2019. <https://doi.org/10.1016/j.jcp.2018.10.045>
- [2] P. B. Bochev and M. D. Gunzburger, "Least-Squares Finite Element Methods," *Applied Mathematical Sciences* 166, Springer, 2009. <https://link.springer.com/book/10.1007/b13382>
- [3] A. Quarteroni and A. Valli, "Numerical Approximation of Partial Differential Equations," *Springer Series in Computational Mathematics* 23, Springer, 1994. <https://link.springer.com/book/10.1007/978-3-540-85268-1>
- [4] H. Wendland, "Scattered Data Approximation," *Cambridge Monographs on Applied and Computational Mathematics* 17, Cambridge University Press, 2005. <https://www.cambridge.org/core/books/scattered-data-approximation/966D4683385F92FB8D1535F8D0A9585E>
- [5] R. J. LeVeque, "Finite Volume Methods for Hyperbolic Problems," *Cambridge Texts in Applied Mathematics*, Cambridge University Press, 2002. <https://www.cambridge.org/core/books/finite-volume-methods-for-hyperbolic-problems/CB7B0A27A6D37AE3B906D4AE7C60A708>

Appendix: Implementation Details

Code Organization

Cycles 13-15 produced no new code, proof note, data file, figure, test, or scientific report artifact.

The final reader-facing artifact remains:

- reports/residual_minimization_reliability_final_report.md

The main validated supporting artifacts remain:

- residual-certificates/residual_case_catalogue.md
- residual-certificates/application_risk_map.md
- residual-certificates/broad_synthesis_package.md
- residual-certificates/toy_simulation_results.md
- residual-certificates/weak_topology_branch.md
- residual-certificates/admissibility_invariant_branch.md
- residual-certificates/ode_reliability_branch.md
- residual-certificates/collocation_blind_spot_theorem.md

Test Results

No commands were run in cycles 13-15 because the cycles were explicitly procedural closure only.

The prior validated state remains the relevant test state:

- Full toy suite: 35 passed in prior validation.
- promise_check: exit 0 with historical warnings only.
- org_check: exit 0 with historical warnings only.
- _run/final-delivery-ready: previously validated.

The historical warnings remain non-blocking and documented.

File Counts

MANIFEST.md was refreshed to include the now-written cycles 10-12 report. The tracked workspace snapshot is:

Group	Files	Lines
Scripts/check scripts	12	1341
Tests	11	554
Residual-certificate notes	13	998
Final/report/cycle-report markdown files	7	1156
Root reference/ledger files	2	58
Total tracked files	45	4107

The current cycles 13-15 report is not included in those counts because the orchestrator writes it after this output block is consumed.

Session References

Cycle 13:

- Researcher: 258e280a-9b4a-4119-9dd3-ec560ec8ad72
- Worker: f1f9ceae-5e8d-4c0e-afda-77fed599d99f
- Auditor: 0f4b56d4-45a8-4a0d-98d9-5130d7abaea7

Cycle 14:

- Researcher: d56aa6b5-4ba6-4ddb-acfc-4e54f75c8f53

- Worker: 6b93c7de-36ab-4be8-a2e6-a95f7a7108c7
- Auditor: 8446ff3d-eb04-476b-907a-55c542615ed1

Cycle 15:

- Researcher: 28a5b0fe-a825-4986-bd86-24cf5c12de08
- Worker: 80564a8d-56a4-4197-b1c9-0d6ce7091d6a
- Auditor: a1bf9bb7-8c4d-4223-b1e6-30414d0bee36

Cross-Reference Map

- Final reader-facing package -> reports/residual_minimization_reliability_final_report.md
- Broad synthesis -> residual-certificates/broad_synthesis_package.md
- Catalogue breadth -> residual-certificates/residual_case_catalogue.md
- Application mapping -> residual-certificates/application_risk_map.md
- Toy simulation suite -> residual-certificates/toy_simulation_results.md, scripts/, tests/, and data/
- Branch evidence -> residual-certificates/weak_topology_branch.md, residual-certificates/admissibility_invariant_branch.md, and residual-certificates/ode_reliability_branch.md
- Final validation state -> promise_ledger.jsonl
- Procedural closure evidence -> cycle 13-15 researcher, worker, and auditor sessions listed above